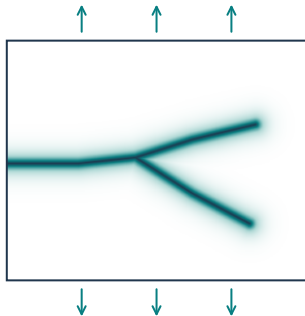


From cracks to computation

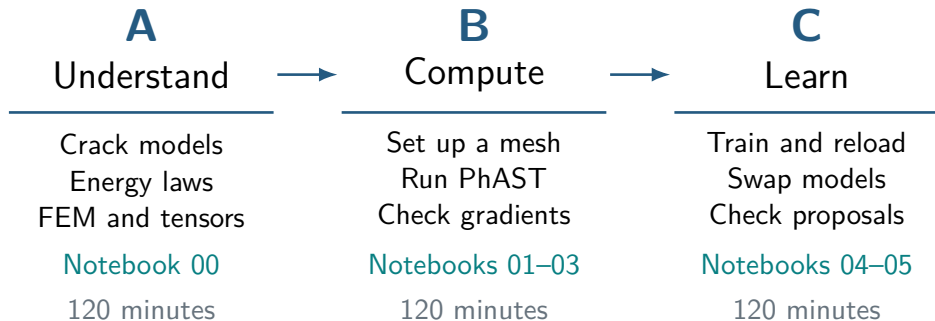
Phase-field fracture, differentiable
FEM and learning with PhAST

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Queen Mary University of London
CEMS-Lab



Phase-field representation (schematic)

Six hours: understand, compute, then learn



Six-hour session design: 330 minutes of teaching and activity, with 30 minutes of breaks.

Understand the crack

Session A / 00–10

Read a damage field.

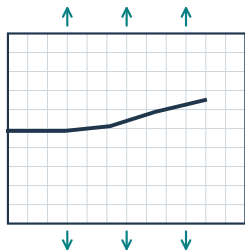
Explain the energy.

Trace one numerical increment.

120 minutes, including a 10-minute break

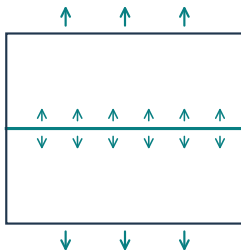
Three descriptions of an evolving crack

Sharp discontinuity



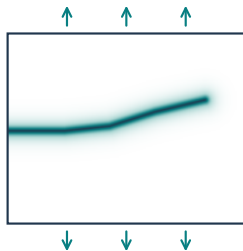
XFEM enriches the approximation around a sharp discontinuity.

Cohesive interface



A cohesive law relates interface traction to separation.

Diffuse phase field



Phase field adds damage d and length scale ℓ .

Original schematics; these approaches can be combined. Miehe, Welschinger & Hofacker (2010), [IJNME 83, 1273–1311](#).

Fracture models and solution algorithms

Question	Examples
What does the crack mean?	Sharp surface, cohesive relation, diffuse field
How is it discretised?	FE spaces, enrichment, mesh, quadrature
How is it solved?	Staggered/monolithic; Newton/quasi-Newton

Quasi-Newton methods solve nonlinear equations within a chosen crack model.

Fracture balances storage and crack cost

A representative small-strain brittle model

$$\mathcal{E}_\ell(u, d) = \underbrace{\int_{\Omega} \left[g(d) \psi^+(\varepsilon(u)) + \psi^-(\varepsilon(u)) \right] dx}_{\text{elastic energy: damage reduces tensile storage}} + \underbrace{\frac{G_c}{c_0} \int_{\Omega} \left[\frac{w(d)}{\ell} + \ell |\nabla d|^2 \right] dx}_{\text{crack cost: local damage + spatial regularisation}} - \underbrace{W_{\text{ext}}}_{\text{loading}}.$$

Francfort & Marigo (1998), [JMPS 46, 1319–1342](#). Bourdin, Francfort & Marigo (2000), [JMPS 48, 797–826](#). Split shown is representative; dynamic mechanics also requires inertia.

AT1 and AT2 choose the crack-density term

$$c_0 = 4 \int_0^1 \sqrt{w(s)} \, ds$$

AT1

$$w(d) = d, \quad c_0 = \frac{8}{3}$$

Finite elastic stage in the standard idealisation.

$$\underbrace{d=0}_{\text{intact}}$$

$$\underbrace{d=1}_{\text{broken}}$$

AT2

$$w(d) = d^2, \quad c_0 = 2$$

Gradual early damage in the standard idealisation.

$$d_{n+1} \geq d_n$$

Degradation $g(d)$ is a separate choice. Pham et al. (2011), [IJDM 20, 618–652](#).

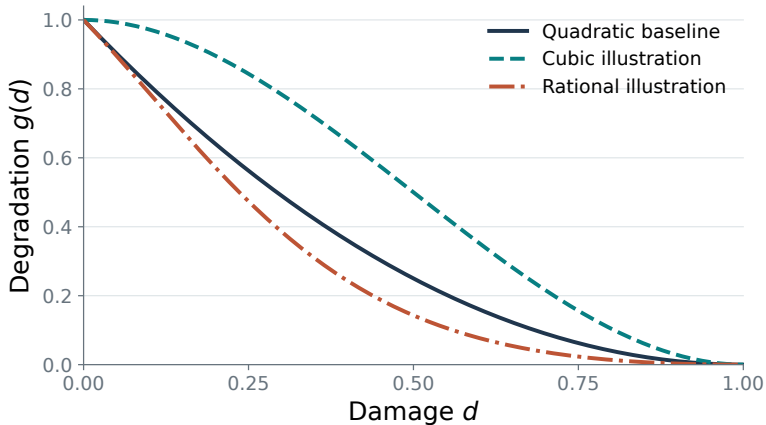
How quickly should stiffness be lost?

Quadratic baseline

$$g(d) = (1 - \eta)(1 - d)^2 + \eta$$

$$g(0) = 1$$

$$g(1) = \eta$$

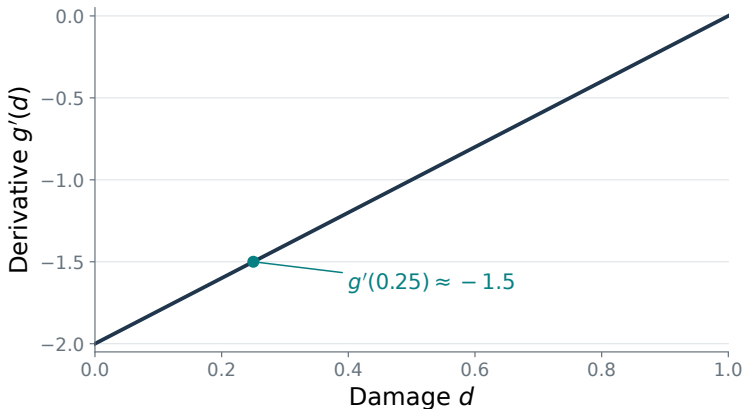


Analytic degradation curves with $\eta = 10^{-6}$; notebooks 01 and 02 use $\eta = 10^{-7}$.

A derivative measures the local energy response

$$g'(d) = -2(1 - \eta) \cdot (1 - d)$$

$$g''(d) = 2(1 - \eta)$$



Analytic sensitivity of the local degradation law.

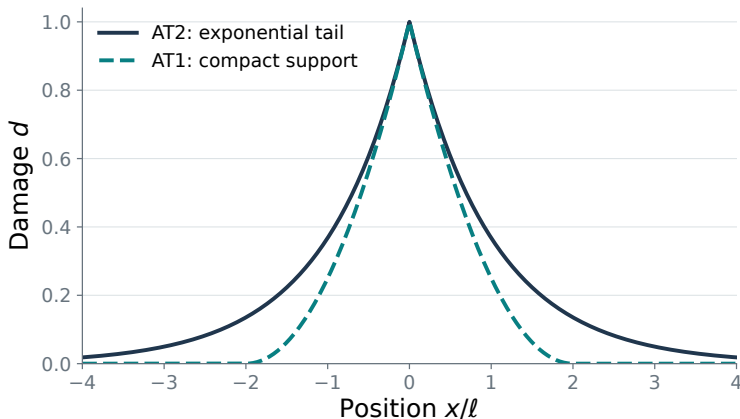
A diffuse crack has a spatial profile

AT2

$$d = e^{-|x|/\ell}$$

AT1

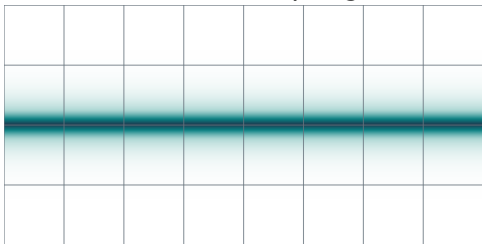
$$d = \left[1 - \frac{|x|}{2\ell}\right]_+^2$$



Isolated one-dimensional crack-density minimisers with $d(0) = 1$. Pham et al. (2011), *IJDM* 20, 618–652.

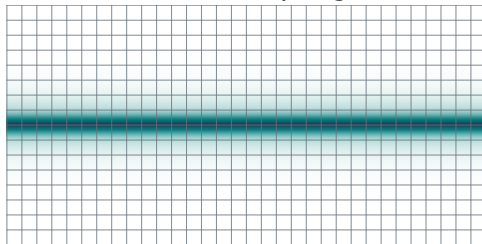
The mesh must resolve the chosen length

Coarse sampling



ℓ : model scale

Finer sampling

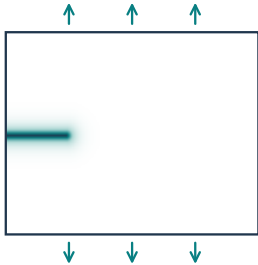


h : discretisation scale

The same analytic damage band on two schematic meshes; resolution is measured relative to ℓ .

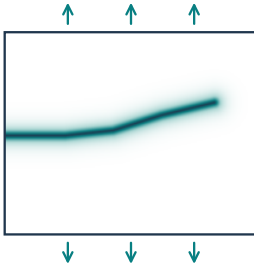
Crack initiation, propagation and branching

Initiation



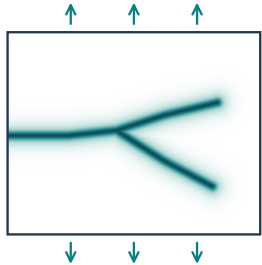
Where does damage first grow?

Propagation



Where does an existing crack advance?

Branching

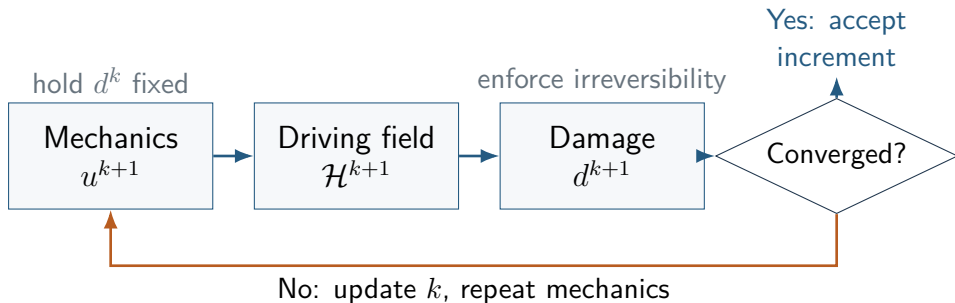


When does one path become two?

Schematic stages of crack initiation, growth and branching.

A staggered step alternates coupled problems

Fix the load increment n ; iterate the coupled fields with index k



History-field notation is representative; irreversibility and the convergence criterion depend on the selected formulation. Miehe, Welschinger & Hofacker (2010), [IJNME 83, 1273–1311](#).

Time integration and nonlinear iteration are different

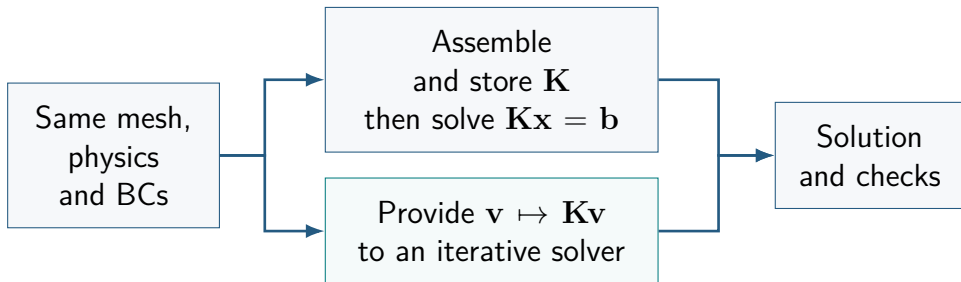
$$\underbrace{\mathbf{M}\ddot{\mathbf{u}}}_{\text{inertia}} + \mathbf{r}_{\text{int}}(\mathbf{u}, d) = \mathbf{f}_{\text{ext}}$$

Static/quasi-static	Equilibrium along a prescribed load path.
Explicit mechanics	Central difference/Verlet; stability limits the time step.
Implicit mechanics	Newmark/generalised- α ; solve an implicit update.
Newton/quasi-Newton	Iterations for a nonlinear algebraic system.

An explicit mechanics update can still use an implicit damage solve. Notebook 01 follows a quasi-static load history.

Matrix-free means applying an operator

$$\mathbf{y} = \mathbf{K}\mathbf{v} \quad \longleftrightarrow \quad \mathbf{y} = \sum_e \mathbf{A}_e^\top \mathbf{K}_e \mathbf{A}_e \mathbf{v}$$



A matrix-free action evaluates the linearised FE operator through local contributions. [PhAST source and documentation](#).

Tensors still have physical locations

Field	Typical shape	Meaning
Coordinates	$N \times n_{\text{dim}}$	positions
Connectivity	$N_e \times n_{\text{loc}}$	local-to-global map
Displacement, damage	$N \times n_{\text{dim}}; N$	nodal fields
Strain, history	element/quadrature axes	local constitutive state

Illustrative tensor layouts; inspect array shapes and ordering in the notebook.

Build, run and differentiate

Session B / 00-04

Specify a problem.

Read its fields and response.

Check a derivative before recovery.

120 minutes, including a 10-minute break

Start from the same computational state

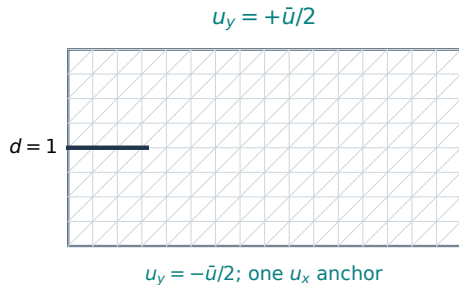
Notebook 01 / setup and version record



PhAST 0.16.2 f6324f899f07

Load the provided course environment and verify the PhAST source revision. [PhAST source and documentation](#).

Geometry, mesh and precrack are separate inputs



Fast route: rectangular T3 mesh from tensors.

The precrack is prescribed $d = 1$ on centreline nodes.

Geometry/import extension: create a Gmsh mesh and inspect named groups.

Original mesh/BC schematic: symmetric opening and a prescribed damage precrack.

Boundary sets and prescribed values



$$\text{top} \longrightarrow \{i_1, i_2, \dots\} \longrightarrow u_y \longrightarrow \bar{u}$$

Inspect group membership and coordinates before solving. The quick tensor-mesh route identifies boundary node sets; imported meshes require their own group checks.

A reproducible calculation has a complete specification

Input	Tiny public PhAST route
Material / fracture	E, ν, G_c, ℓ, η ; AT2, isotropic degradation
Mesh / precision	T3; 128×64 cells; float64
Loading	60 quasi-static increments; symmetric opening
Numerics	SciPy mechanics; damage/outer tolerances; failure checks

Quasistatic AT2 with assembled sparse-direct mechanics. Case values: `configs/day2_forward/tiny_notched_tension.json`.

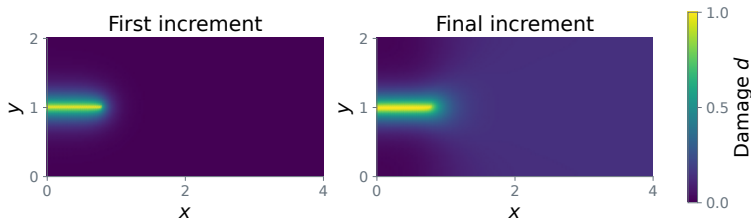
Distinguish the seeded crack from new damage

AT2 crack cost

$$\gamma_\ell(d) = \frac{d^2}{2\ell} + \frac{\ell}{2} |\nabla d|^2$$

$$G_c \int_{\Omega} \gamma_\ell(d) \, dx$$

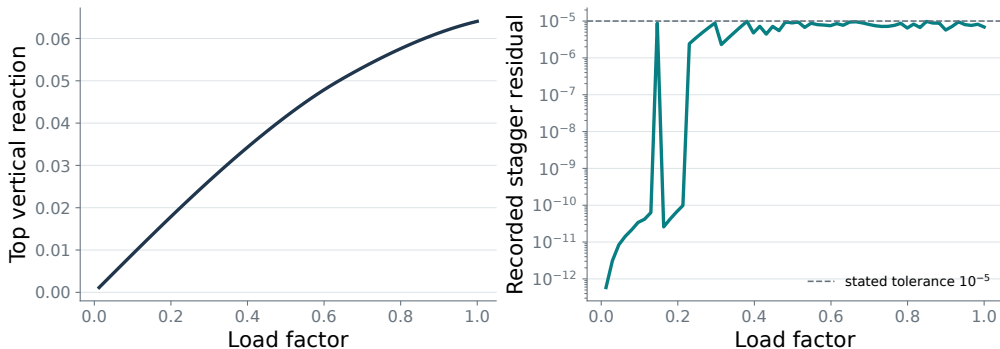
Seed: prescribed $d = 1$



History: $d_{n+1} \geq d_n$

PhAST quasistatic seeded-crack response over 60 increments; source f6324f899f07.

Read response and numerical checks together



Residuals assess nonlinear solve convergence; assess mesh convergence and physical agreement separately.

Change one input, then explain the difference

Five-minute comparison / followed by a ten-minute break



A material/load change or a numerical refinement

Changing ℓ changes the regularised model; changing h refines its discretisation.

Tensor code can mirror the finite-element calculation

```
u_e = u[connectivity]
strain_q = B @ u_e
stress_q = law(strain_q, d_q)
r_e = integrate(B.T @ stress_q)
r = accumulate(r_e, connectivity)
```

Gather

Local physics

Weighted quadrature

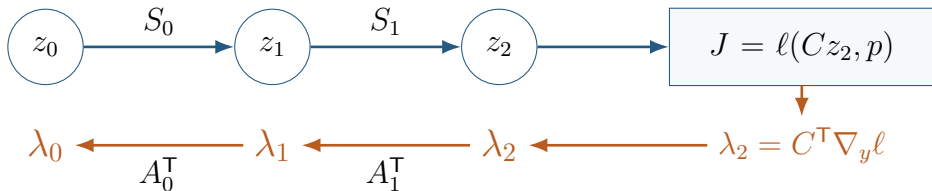
Global accumulation

Element-level pseudocode; batched implementations combine the same local operations.

A gradient follows a specific computational path

$$z_{n+1} = S_n(z_n, p); A_n = \partial S_n / \partial z_n, B_n = \partial S_n / \partial p$$

the same p enters each update



$$\frac{dJ}{dp} = \underbrace{B_0^T \lambda_1 + B_1^T \lambda_2}_{\text{add every use of } p} + \underbrace{\partial_p \ell + (\partial_p z_0)^T \lambda_0}_{\text{direct and initial-state dependence}}$$

Custom solves, convergence, boundary treatment, detach, clamps, history maxima and active sets matter. [PyTorch autograd documentation](#).

Check a local derivative three ways

Notebook 02 / analytic, automatic, finite difference

$$g'(d) = -2(1 - \eta)(1 - d) \quad D_{\epsilon}g(d) = \frac{g(d + \epsilon) - g(d - \epsilon)}{2\epsilon}$$

```
d = torch.tensor(0.25, dtype=torch.float64,  
                 requires_grad=True)  
(g_ad,) = torch.autograd.grad(material.degradation(d), d)
```

PhAST local degradation law: $\eta = 10^{-7}$, $g'(0.25) = -1.49999985$.

Choose an informative inverse observation

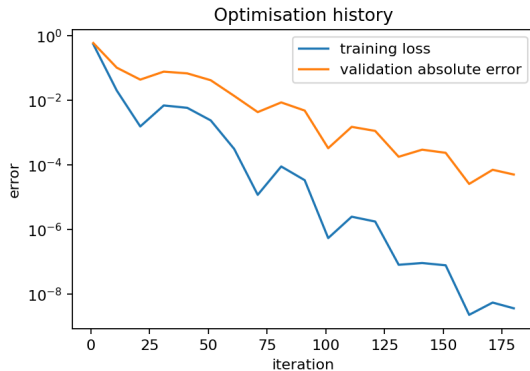
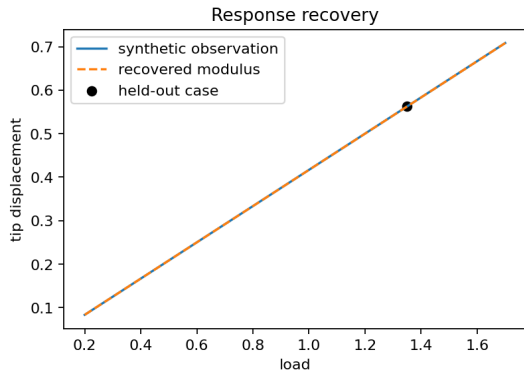
Notebook 03 / elastic-bar parameter recovery

$$u_{\text{tip}}(E) = \frac{FL}{EA} \quad \mathcal{L}(E) = \frac{1}{2} \sum_j \left[u_{\text{tip}}(E; F_j) - u_j^{\text{obs}} \right]^2$$



Fixed-mesh, one-parameter recovery; optimising $\log E$ preserves positivity.

A small loss is only one part of recovery evidence



Notebook 03: parameter and response recovery for the elastic bar.

Keep an experiment someone else can reproduce

1. Configuration and source version.
2. Whole field, response and numerical checks.
3. Derivative check and inverse interpretation.

Session B hand-in: one executed notebook and one comparison sheet, including precision, hardware and measured timing scope.

Train, save and assess a model

Session C / 00-04

Fit a compact model.

Reload the same prediction.

Assess a proposal and retain correction.

120 minutes, including a 10-minute break

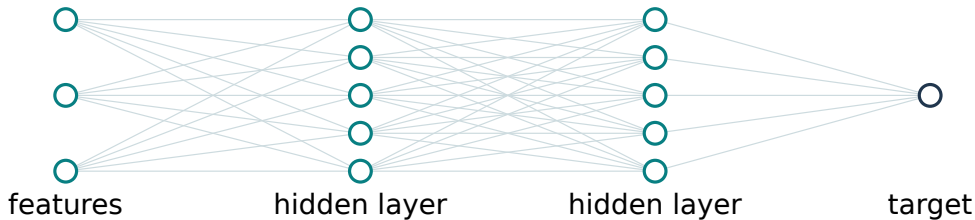
The data card defines what learning means

Notebook 04 / course-owned field equation

Item	Teaching example
Input	Nodewise $[x/L, y/H, \text{load factor}]$
Target	A scalar damage-like teaching field
Split	Whole load cases: train, validation, held-out test
Provenance	Discrete Helmholtz-like reference equation

Separate whole cases/trajectories to prevent leakage. Feature order, units, location and normalisation are part of the model contract.

An MLP maps named features to a named target



$$\hat{d} = \mathcal{N}_{\omega}(\text{normalise}(x/L, y/H, \lambda))$$

A pointwise model uses local input features; graph models additionally encode connectivity.

A fitted model can still miss the local field

Toy reference

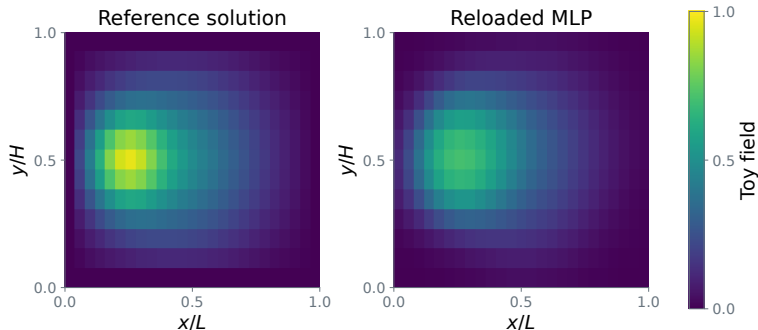
$$(\mathbf{I} + \ell^2 \mathbf{L}_h) d = b$$

Model

$$\hat{d} = \mathcal{N}_\omega(x/L, y/H, \lambda)$$

Same held-out load.

Same colour scale.



Notebook 04: Helmholtz-like field prediction at held-out load $\lambda = 1$.

A checkpoint is more than weights

Notebook 04 / save, new instance, reload, compare

```
checkpoint = {  
    "state_dict": model.state_dict(),  
    "architecture": configuration,  
    "features": feature_order,  
    "normalisation": scaling,  
    "source": data_provenance,  
}
```

$$\max_i \left| \hat{d}_i^{\text{before}} - \hat{d}_i^{\text{reload}} \right| \leq \tau$$

Illustrative checkpoint layout; use the notebook's exact schema. Preserve shape, precision, evaluation mode and supported scope. [PyTorch autograd documentation](#).

A model swap requires compatible inputs

Representation	Required input construction
MLP / RBF	Named point features; scaling; RBF centres/kernel
GNN / GNO	Nodes, edges/neighbourhoods and geometry/operator data
CNN	Grid/channel order and explicit mesh-grid maps

The MLP and RBF share one field interface. Mesh transfer requires compatible geometry and discretisation inputs.

A learned model supplies a proposal

Notebook 05 / a defined teaching adapter



Detached inference and differentiable training use distinct computation graphs

Notebook 05: model inference followed by residual assessment for the teaching equation.

Admissibility and equilibrium

$$d^{\text{proj}} = \max\left(d^{\text{previous}}, \min(1, \max(0, \hat{d}))\right)$$

Projection can enforce

$$0 \leq d \leq 1$$

$$d \geq d^{\text{previous}}$$

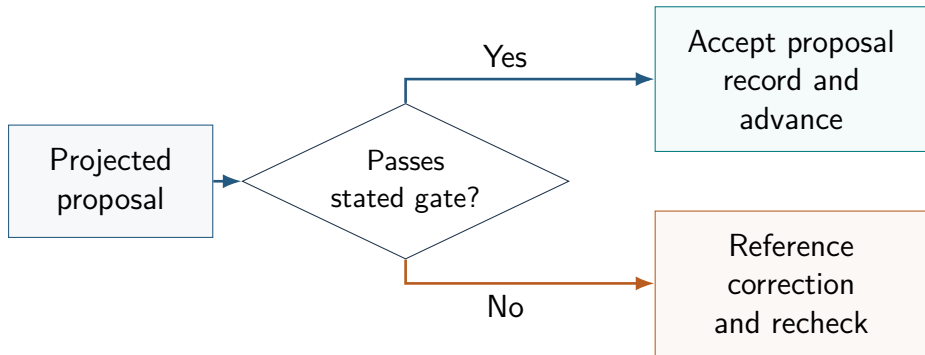
Assessment still asks

Are the equations satisfied?

Are prescribed values consistent?

Assess admissibility, residual convergence and energy consistency with criteria appropriate to the model.

Acceptance needs an explicit correction route



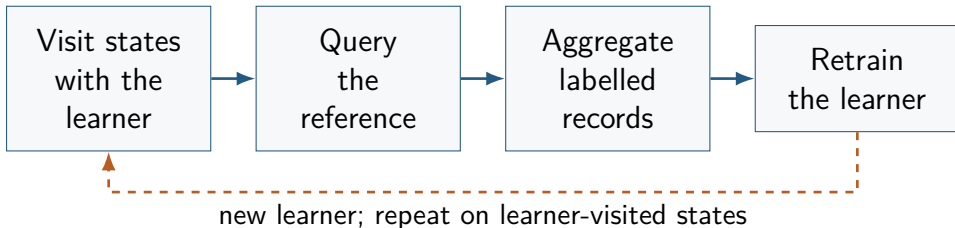
If correction also fails: stop and diagnose.

Record the proposal, projection, residual, decision, correction and total computational cost.

DAgger learns from states the model visits

Conceptual extension: the current notebook saves one replay record

Keep held-out evaluation separate from the aggregated training data.



Ross, Gordon & Bagnell (2011), [AISTATS: dataset aggregation](#). Inspect the saved replay record and design one data-aggregation round.

Architectures encode different assumptions

Point map	Compact features; explicit coordinate assumptions
Grid or graph	Structured stencil or irregular neighbourhood
Operator map	Function inputs/outputs; training and discretisation scope

Choose a representation for the input structure, discretisation and intended evaluation setting.

Read a research example through the same evidence

1. Whole field and physical load/time history.
2. Matched response, numerical checks and held-out scope.
3. Source/configuration and complete computational cost.

Research discussion: compare the physical problem, numerical method, accuracy and computational cost.

A useful contribution starts with reproducibility



github.com/CEMS-Lab/PhAST

Read current contribution and issue guidance before submitting. Keep private data, credentials and unpublished results out of public reports.

Keep the chain from assumptions to evidence visible

A Representation $\rightarrow$ energy $\rightarrow$ discretisation

B Configuration $\rightarrow$ result $\rightarrow$ derivative

C Data $\rightarrow$ prediction $\rightarrow$ assessment

References and further reading: [slides/README.md](#) and the companion book. Original course graphics.